JIAMING YANG

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Computer Science and Engineering, University of Michigan

RESEARCH INTERESTS

I study **randomized numerical linear algebra (RandNLA)**, with applications in large-scale optimization problems and machine learning. I am broadly interested in machine learning theory, and the intersection of randomized algorithms, random matrix theory and statistics.

EDUCATION

Computer Science and Engineering (CSE), University of Michigan, Ann Arbor *08/2022 - Present*
Ph.D. Candidate. GPA: 4.0.

School of Mathematical Sciences (SMS), Peking University *09/2018 - 07/2022*
B.S. in Statistics.

RESEARCH EXPERIENCE

University of Michigan, Ann Arbor *08/2022 - Present*
Graduate Student Research Assistant. Advisor: Prof. Michał Dereziński

- Working on randomized numerical linear algebra and machine learning theory.

Peking University *12/2021 - 06/2022*
Undergrad thesis. Supervisor: Prof. Hao Ge

- Established a statistical characterization of a single cell RNA-seq technique. Designed a hypothesis testing framework for judging whether the expression levels of a specific gene in two single cells are the same.

Adobe Research (Remote) *06/2021 - 06/2022*
Supervisor: Dr. Zhao Song

- Worked on the theoretical foundations of machine learning (sparse training scheme, over-parameterized networks, adversarial training, etc.). Resulted in two papers in ICLR and ICML.

Columbia University (Remote) *06/2021 - 12/2021*
Supervisor: Prof. Josh Alman

- Designed a recursive algorithm for solving Kronecker product regression problem exactly and proved its optimality. Applied the result to Kronecker matrix low-rank approximation problem solving.

PUBLICATIONS

[1] **"Randomized Kaczmarz Methods with Beyond-Krylov Convergence",**
M. Dereziński, E. Rebrova, D. Needell, **J. Yang** *(α-β)*, **SIAM Journal on Matrix Analysis and Applications**.

[2] **"Have ASkotch: A Neat Solution for Large-scale Kernel Ridge Regression",**
P. Rathore, Z. Frangella, **J. Yang**, M. Dereziński, M. Udell, Journal of Machine Learning Research (**JMLR**).

[3] **"Faster Linear Systems and Matrix Norm Approximation via Multi-level Sketched Preconditioning",**
M. Dereziński, C. Musco, **J. Yang** *(α-β)*, Symposium on Discrete Algorithms (**SODA 2025**).

[4] **"Solving Dense Linear Systems Faster than via Preconditioning",**
M. Dereziński, **J. Yang** *(α-β)*, Symposium on Theory of Computing (**STOC 2024**).

[5] **"HERTA: A High-Efficiency and Rigorous Training Algorithm for Unfolded Graph Neural Networks",**
Y. Yang, **J. Yang**, W. Hu, M. Dereziński, Mathematics of Modern Machine Learning (**M3L, NeurIPS 2024**).

[6] **"Federated Adversarial Learning: A Framework with Convergence Analysis",**
X. Li, Z. Song, **J. Yang** *(α-β)*, International Conference on Machine Learning (**ICML 2023**).

[7] **"Pixelated butterfly: Simple and efficient sparse training for neural network models",**
B. Chen, T. Dao, K. Liang, **J. Yang**, Z. Song, A. Rudra, C. Ré,
International Conference on Learning Representations (**ICLR 2022**), **spotlight**.

WORKING PAPERS

[1] **"Accelerating Randomized Subspace Newton via Regularized Projections",**
J. Yang, P.L. Poirion, A. Takeda, M. Dereziński. In submission to **ICLR**.

SKILLS

Programming: Python, Matlab, R, LaTeX.

Languages: Native in Chinese; Proficient in English (GRE $159 + 170 + 4.0$, TOEFL 107 with speaking 25).

TALKS

INFORMS 2026 (San Francisco): Faster Randomized Subspace Newton via Nesterov Acceleration. *11/2026*
ICCOPT 2025 (Los Angeles): Randomized Kaczmarz Methods with Beyond-Krylov Convergence. *07/2025*
SODA 2025 (New Orleans): Faster Linear Systems via Multi-level Sketched Preconditioning. *01/2025*
UMich Theory Lunch: Solving Dense Linear Systems Faster than via Preconditioning. *03/2024*

SERVICES

Instructor for EECS 376: Foundations of Computer Science (Winter 2026).
Reviewer of NeurIPS 2024, 2025, ICML 2025, 2026, AISTATS 2025, 2026.
Reviewer of UMich CSE PhD program applicants.
UMich CSEG Buddy Program: advisor of two graduate students.

AWARDS AND HONORS

Department finalist of Two Sigma PhD Fellowship *2025*
RMT+NLA II Travel Grant *2025*
SODA Student Travel Grant *2025*
STOC Student Travel Grant *2024*
University of Michigan Rackham Conference Travel Grant *2023, 2024, 2025, 2026*